An La

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Education

2021 - present MS/PhD (Computer Science) at **University of Massachusetts Amherst** (GPA: 3.80/4.0)
Selective taken courses: Algorithms with Predictions, Randomized Algorithms, Algorithms for Data Science, Probabilistic Graphical Models, Distributed and Operating Systems.

2013 - 2017 Bachelor's Degree at **Honors Program, VNU-HCMUS, Vietnam** (GPA: 3.57/4.0)
Information Technology, Graduated with distinction.
Thesis: *Multi-modal video retrieval using Dilated Pyramidal Residual network*.

Skills

Theory: Design data structures & algorithms, statistics and probability, quantitative analysis and modeling.

Programming: Proficient in Python to build machine learning models (TensorFlow, PyTorch, Scikit-Learn), process, analyze and visualize data (Numpy, Pandas, Seaborn).
Proficient in C++ to implement algorithms in competitive programming contests and image processing projects with OpenCV (text region extraction, image enhancement).

Experience

Sep - Dec, 2025: Applied Scientist Intern – Recommender Systems / Information Retrieval at Amazon

- ⋆ Design an explainable exploratory process for time-series forecasting problems.
- ⋆ Design an exploratory assistant for data scientists to investigate statistical models.
- ⋆ Technical skills: *forecasting design*, *prompt engineering*, *LLMs applications*.

July - August, 2025: Data Scientist Intern at GeoComply

- ⋆ Designed and implemented a dynamic clustering framework for robust classification problems.
- ⋆ Designed and developed a multi-agent system for fraud detection.
- ⋆ Improved time efficiency by 4x and discovered new fraud patterns.
- ⋆ Technical skills: *data structure design*, *agentic-system design*, *LLMs applications*.

2021 - now: Research and Teaching Assistant at Theory CS Group/UMass Amherst

- ⋆ Study data structures/algorithms in computational geometry and their applications in data mining.
- ⋆ Study online algorithms with machine learning predictions.
- ⋆ Teaching Assistant: Algorithms for Data Science (Fall22, Spring25), Advanced Algorithms (Spring23).
- ⋆ Publications: *Dynamic Locality Sensitive Orderings in Doubling Metrics* in STOC'25, *New weighted additive spanners*, and Forecasting Time Series with LLMs to be appeared in PACLIC 39.
- ⋆ Technical skills: *algorithm design*, *data structure design*, *algorithmic analysis*.

2020 - 2021: Data Scientist at PrimeData, Vietnam github/anla11/analytic_marketing

- ⋆ Designed and implemented an automatic analysis framework for algorithmic marketing applications.
- ⋆ Technical skills: *quantitative analysis*, *Bayesian machine learning* and *probabilistic programming*.

2017 - 2019: Data Scientist at FPT Telecom, FPT Group, Vietnam github/anla11/adaptive_cf_recsys

- ⋆ Designed and implemented a graph-based model with multiple evaluation metrics for a recommender system.
- ⋆ Increased precision by 6% while maintaining diversity, coverage, and congestion.
- ⋆ Publication: *Adaptive Collaborative Filtering for Recommender System* in ICCS 2019.
- ⋆ Technical skills: *content-based, collaborative-filtering, and graph-based techniques.*

Selective Awards

Dec. 2016	National Vietnam award for Outstanding Female Students in Science and Technology
Aug. 2016	Awards from Facebook Hackathon Vietnam 2016
	1^{st} prize of Most Innovative Product
	2^{nd} prize of Best Product in Facebook Marketing Category
2014	*2^{nd} prize* in ACM-ICPC Vietnam National 1^{st} Round
2013	*3^{rd} prize* in Informatics at the Vietnam National Excellent Student Exam